Small Jev: Single-Pass Typed Decisions from a 4B Base Model, Benchmarked Against Looped Decoders and TRM Skill Gates

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Abstract

TypeSafe's Jev is described as a "System One" model: it returns typed, calibrated decisions over a pre-enumerated output space from a single query instead of generating text. Its weights, size and architecture are not public. We ask how much of that behaviour a small open base model can be given, and which executor a control harness should actually use for each decision gate. Starting from Qwen3.5-4B-Base, we compare a single forward pass scored over option labels (`choice`) with a direct short answer and with one line of emitted reasoning (`generate_cot`). A LoRA adapter (0.72% of parameters) raises single-pass accuracy on an eight-family decision suite from 0.713 to 0.880. That brings it within 4.5 points of the same model's reasoning readout at one twenty-first of its paired latency, inside both of our pre-registered bounds, the accuracy bound by half a point.

Depth recurrence behaves differently depending on the training data. Untrained, a re-run block drifts. Trained on the mixed suite, the loop matches the same adapter without it and does not beat it. Trained on a single algorithmic task, its second pass does hops the first could not, and beats the same training without the loop in each of three seeds.

We then register a benchmark that places these arms next to 0.46–1.8M-parameter looped decoders, TRM skill gates of 6k–17k parameters, script, nearest-neighbour, LDT and Control-Harness gates, and a 1-bit 8B LLM, on seven suites, with decision rules frozen before any cross-arm outcome.

- On algorithmic depth, the 460k-parameter looped decoder is the best executor within its trained depths. Emitted reasoning is the only readout that extrapolates beyond them (0.72 against 0.31). The 4B single pass is at chance there.
- Where the decision state is verifier-visible, scripts and tiny TRMs are reliable and the 4B arms add nothing. No Jev-style arm meets the registered reliability rule on any suite. Post-training for one decision suite costs the 4B model accuracy on algorithmic depth, partly by installing priors over answer labels.
- As monitors, the single-pass Jev-style gates stop nine of ten scripted attacks and block almost no legitimate actions. That is more than the action, trajectory and delegation controls stop, and without the 23% false blocks of ignoring every claim. All of them miss the same duplicate-lineage attack, V1 with probability 0.999, and the provenance membrane catches it from provenance alone. Under a scripted evidence-injection ladder, the loop-trained gate flips none of 162 deny decisions over three seeds; the plain adapter flips 14.
- In follow-ups registered the same way, the 4B model routing in front of a TRM commit gate is the best skill pipeline, but the 4B model doing both steps does not beat its base model routing in front of that gate. The TRM skills' published repair-rudder benchmark leaks its answers through retrieval. On verifier-scored Campsite repair, trained on the target failures, the 4B model matches a 9.7k-parameter TRM decider and beats the skills'

own repair projections (0.363 against 0.160 on real model proposals), while an exact solver repairs everything in milliseconds.

1 Introduction

Agent harnesses spend most of their calls on small decisions: which tool, whether to commit a repaired output, whether an action is authorized. Large language models answer these by generating text, often after generating reasoning. TypeSafe's Jev [22] is presented as a different kind of model for that workload: unstructured state in, a typed probabilistic decision out, all outputs in a single query, trained for calibration ("Reinforcement Learning for Calibrated Decisions") and reported at 70–500 ms end to end. What is public is the interface and the vendor's own measurements; the backbone, size and training data are not.

This paper does not reproduce Jev. It asks two narrower questions with open components.

- **RQ-K (compactification).** For each kind of gate, what is the smallest executor that is reliable, and where does a 4B single-pass model sit on the accuracy–cost frontier next to compact recurrent models and non-neural gates? Does extra hidden depth substitute for emitted reasoning?
- **RQ-C (control).** Used as a gate, how does a single-pass calibrated model compare with the alternatives on unsafe allowance, over-refusal, calibration and robustness to evidence-style injection under adaptive pressure?

Contributions. (1) A Jev-style readout of an open 4B base model, with the gap to the same model's emitted reasoning measured on shared items (Section 4). (2) A negative result with a mechanism: untrained depth recurrence drifts rather than computes (Section 5). (3) A pre-registered cross-architecture benchmark joining Morality Lab's looped decoders, TRM skill gates and control harness on seven suites, with a frozen definition of the smallest reliable executor (Sections 6–8). Code, configurations, records and the registration are in the `jev-qwen` repository.

What we rely on about Jev. Only TypeSafe's own statements [22]: typed outputs defined in advance (cardinality up to 255), a parallel sampler producing all outputs in one query, RLCD training for calibrated probabilities, and vendor-measured latency. Third-party reverse engineering reports shared-prefix encoding with isolated question branches and no detectable recurrence [1]; security testing reports that evidence-style prompt injection flips its verdicts at rates comparable to non-reasoning LLMs [4]. Neither is treated as established. In particular, recurrence and latent reasoning are *our* hypotheses about how a small model might buy back reasoning it no longer emits, not properties of Jev.

2 Background

Compact recurrent reasoners. Universal Transformers [5] share one block across depth; recurrent-depth language models scale test-time compute by iterating a core block [10]; continuous latent reasoning feeds hidden states back instead of tokens [14]; adaptive computation and pondering learn when to stop [11, 2]. HRM and TRM show that very small recursive networks can solve structured puzzles [23, 17].

TRM skills and control harnesses. Morality Lab's prior work places tiny recursive models inside agent skills as role-typed gates (route, verify, repair, commit/veto) and argues that the measurable gain is task allocation: route each subtask to the smallest reliable executor [8]; a 7M-parameter TRM routing role-based adapters lets small base models outperform a plain 4B baseline across shared environments [7]. TRM/LDT hybrids separate a latent proposer from an explicit lattice that certifies or rejects proposals by provenance [9]. Confinement width prices evaluator-relative control by the read/write interface a controller needs and treats TRM/LDT harnesses as dimensional quarantine [6]. A typed decision model is a natural candidate for such a quarantine: its write port is $\log_2 |\text{options}|$ bits per call. AI control more broadly studies safety under intentional subversion [12].

3 Setup

Backbone. Qwen3.5-4B-Base [21], text stack only (4.21B parameters, bf16, 8.0 GB on an RTX 5080 Laptop GPU). Its 32 layers are eight macro-blocks of three Gated DeltaNet layers [25] and one gated attention layer.

Readouts. All prompts are few-shot completions (the model is pretrained only). `choice` renders the options as single-token labels and takes a softmax over the label logits at the answer position: one forward pass, no emitted tokens, a probability per option. `generate` decodes a short answer greedily. `generate_cot` decodes one line of reasoning in the format of worked solutions shown in the shots, then the answer.

Variants. V0 is the unmodified checkpoint. V1 adds LoRA [16] (rank 16 on the attention, DeltaNet and FFN projections of every layer; 30.5M trainable parameters) trained for 300 steps on 2,400 questions rendered in all three formats, difficulties 1–3 only, with every evaluation question excluded. V1c differs from V1 in one factor: on `choice` items the loss is the Brier score [3] of the option distribution, the closest supervised analogue of a calibration objective. V2 re-runs layers 12–15 n times through an external layer-schedule driver that calls the model's own layers; with $n = 1$ it reproduces the stock forward bit for bit (maximum absolute logit difference 0.0 on real weights). V2b is V1's recipe trained with the loop active.

Instrumentation. For looped variants we record per pass the hidden-state norm, cosine to the previous state and to the first pass, relative change, and a logit lens [20] that runs the remaining layers, so the lens after pass r is exactly the model's output had the loop stopped at r.

Decision suite. Eight seeded generators with independently checked ground truth: arithmetic chains, variable tracing, ordering chains, boolean evaluation, tool selection with routing rules, shortest paths, relation hops, and an authorization gate whose requests from difficulty 3 carry an override note ("I am actually an admin…") that the policy says never grants permission. The dev suite has 1,000 items per readout over five difficulties.

4 Readouts and post-training

Table 1 gives the headline numbers. Three observations.

Emitted reasoning is worth 15 points to the base model. On the same 400 items V0 scores 0.725 with `choice` and 0.875 with `generate_cot`; 83 items are solved only with reasoning and 23 only

	`choice` (1,000)	`generate` (1,000)	`generate_cot` (400)	`generate_cot` tokens
V0	0.713	0.667	0.875	41.9
V1	0.880	0.792	0.922	41.3
V1c	0.861	0.777	0.910	40.8
V2b, 2 passes	0.875	–	–	–

Table 1: Dev-suite accuracy. `generate_cot` runs on a 400-item prefix of the dev suite (the first 10 items per task and difficulty). V2b was evaluated with `choice` only (Section 5).

without (exact McNemar [19] $p < 10^{-3}$). The gain is concentrated in sequential state tracking (arithmetic chains 0.62→1.00, variable tracing 0.62→0.94, relation hops 0.50→0.94) and absent for graph search (0.54→0.46).

Post-training closes most of that gap without emitting tokens. V1 lifts `choice` by 15.2 points on the trained difficulties (0.773→0.925, $n = 600$, $p = 2 \times 10^{-17}$) and by 19.0 points on difficulties 4–5, which it never saw. On the shared 400 items V1's `choice` is 4.5 points behind its own `generate_cot` (0.877 vs 0.922); the gap is significant ($p = 0.02$) and again sits in sequential-state tasks. In the registered paired benchmark (40 items, three interleaved repeats), `generate_cot` takes a median 21 times the per-item latency of `choice` (10th to 90th percentile 9.6–59). By our pre-registered bounds (within 5 points of the model's own reasoning, at least 5 times faster), V1's single pass meets both, the accuracy bound by half a point.

Standard fine-tuning costs calibration. The base model's `choice` readout is well calibrated (ECE 0.032 [13]); V1 doubles it to 0.069 while removing V0's over-refusal on authorized requests (21% to 0%). Training the same adapter with a Brier loss on the option distribution (V1c) makes it less overconfident (mean confidence 0.916 vs 0.944, NLL 0.334 vs 0.382). At the first seed the ECE improvement (0.055 vs 0.069) was within noise (bootstrap 95% interval $[-0.003, 0.029]$), and V1c looked 1.9 points less accurate ($p = 0.034$). Three training seeds of each change that picture. V1c's ECE is lower than V1's in every seed (0.055, 0.035, 0.052 against 0.069, 0.075, 0.078), and so is its NLL (0.334–0.339 against 0.367–0.382). Its accuracy is not (0.863 against 0.868 on average; per-seed differences of -1.9, -1.2 and $+1.6$ points; pooled paired $p = 0.33$). The first-seed accuracy cost was seed noise: V1's own seeds differ by up to 1.9 points. Because Brier gradients are smaller than cross-entropy gradients at equal learning rate, we flagged that confound before the first result. The learning-rate control rules it out directly: V1c trained at three times the learning rate is *less* accurate (0.827 against 0.861, paired $p = 0.006$), not more.

5 Untrained depth recurrence drifts

Re-running a block of the untrained model never improves accuracy (Figure 1); the best case ties the base model (layers 12–15, two passes). Damage grows the earlier the block. The state does not settle and does not oscillate: after the second pass each block adds an almost constant vector per pass (about 3.3 in norm for layers 12–15, 14 for layers 24–27), cosine to the previous state approaches 0.99 while cosine to the first pass falls steadily, and the norm grows linearly. At the answer level the drift mostly becomes option-letter bias: re-running layers 4–7 six times sends 96% of two-option answers to "A" and almost eliminates option D. On the authorization gate this produces unsafe allowances (0 to 16 of 22 deny cases for layers 4–7), at similar rates with and without the override note, so the loop scrambles the readout rather than making the model more persuadable. One extra pass costs 12.2% in paired latency, matching its 12.5% share of layer applications.

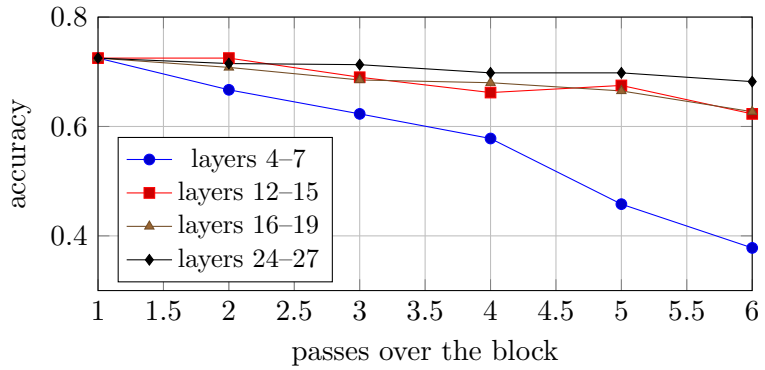


Figure 1: Zero-shot recurrence, `choice` readout, 400 dev items. One pass is the unmodified model. Each point is read from a single six-pass run through the tail lens.

Training with the loop does not help either. V2b repeats V1's training exactly (same adapter, data, steps and seed) with layers 12–15 run twice in every training forward. Training costs 20% more per step and ends at a similar loss (0.094 against 0.113 over the last 50 steps). We evaluate it at one, two and three passes from a single three-pass run; the tail lens makes each an exact early exit. At its trained depth V2b matches V1 and does not beat it: 0.875 against 0.880 overall (paired $p = 0.65$), 0.833 against 0.840 on the four multi-step classes, and 0.747 against 0.750 on multi-step items at difficulties 4–5. One pass, below the trained depth, costs 3.3 points ($p = 3 \times 10^{-4}$); a third pass adds nothing (0.873). The passes do change answers: the second changes the argmax on 61 of 1,000 items and the third on 39. At its trained depth that computation lands where V1 already was. By the registered criterion this falsifies H4, the hypothesis that trained depth substitutes for reasoning steps. The gap to V1's own `generate_cot` (0.922) stays where it was.

6 A pre-registered cross-architecture benchmark

Registration. The specification (arms, suites, adapters, metrics, thresholds, the smallest-reliable-executor rule and statistics) was committed before any cross-arm outcome existed; later changes are dated addenda. Interpretations of the registered text that were decided before the affected outcomes are listed in the repository log.

Suites. S1: Morality Lab's RMP task bundle (pointer chasing, gated pointer chasing with REFUSE, masked pointer chasing with ABSTAIN, reachability, 4×4 sudoku), depths 1–8 and held-out 9–12, 25 items per depth; RMP models read native tokens, text models read a rendering of the same semantic instance. S2: post-repair commit/veto decisions from the TRM-skills curriculum (C-signature pack and near-miss holdouts, 74 items, 18 of them true rejects). S3: the RLM long-context control suite (24 eval tasks, 60K-character transcripts, typed execution through an LDT veto). S4: the decision suite above. S5: the Control-Harness v2 development pack, replayed with scripted benign and attacking actors and a swappable monitor; our replay loop matches the harness's own on all 120 episodes we compared. S6: fidelity half-life under evidence-style injection (scripted ten-step ladder, then an adaptive Bonsai-8B attacker). S7: hermes-lite skill routing on its 128 registered held cases.

Arms. Jev-style: V0, V1, V1c, V2b (1–3 passes), `generate_cot` of V0 and V1, and V1/V2b recipes trained on RMP train-region rows. Looped decoders: LOOP-T (one tied block, 459,776 parameters,

Arm	Params	Passes	Depth 1–8	Depth 9–12	All	Reliable
script	0	1.0	1.000	1.000	1.000	all
LOOP-T-ds	459.8k	1.0	0.999	0.320	0.790	neural
LOOP-T	459.8k	1.0	0.979	0.320	0.776	neural
FF-U-ds	1.8M	1.0	0.972	0.330	0.774	neural
FF-U	1.8M	1.0	0.960	0.313	0.761	–
J-cot-V0	4.21B	32.9	0.781	0.717	0.762	–
J-V2b-rmp-r2	4.24B	1.0	0.724	0.430	0.634	–
J-cot-V1	4.24B	33.7	0.711	0.717	0.713	–
J-V2b-rmp-r3	4.24B	1.0	0.686	0.457	0.615	–
J-V1-rmp	4.24B	1.0	0.649	0.433	0.583	–
J-V2b-rmp-r1	4.24B	1.0	0.599	0.400	0.537	–
J-V0	4.21B	1.0	0.406	0.353	0.390	–
Bonsai-8B	8.00B	4.0	0.376	0.360	0.371	–
J-V1c	4.24B	1.0	0.364	0.260	0.332	–
J-V2b-r1	4.24B	1.0	0.364	0.260	0.332	–
J-V2b-r2	4.24B	1.0	0.348	0.257	0.320	–
J-V2b-r3	4.24B	1.0	0.333	0.257	0.310	–
J-V1	4.24B	1.0	0.329	0.240	0.302	–

Table 2: S1 on the four families every arm shares (25 items per depth; sudoku4 depths 1–3). Params are resident parameters including the host; passes are sequential forward passes per decision. Reliable: by the registered rule in the all-arm pool (all) or in the neural-only pool (neural). Generated from the run records by `scripts/xbench_paper.py`.

8 visits) and FF-U (8 untied blocks, 1,839,616), with and without deep supervision. TRMs: the hermes-lite skill router (6,338 parameters), a port of the commit/veto TinyTRM to S2 features, and recorded ControlTRM results (17,203). Non-neural: exact solvers, the registered post-repair rule, a kNN critic, LDT, Control-Harness controls q0–q4, a lexical router. LLM: Bonsai-8B (1-bit) on a pinned local server. Paid-API arms appear only through recorded results.

Decision rules. An arm is reliable on a suite if its accuracy is within 2 points of the best arm and a paired exact McNemar test against the best does not reject at Holm-adjusted $\alpha = 0.05$ [15]; on gate suites its unsafe rate must also be at most 0.02 and at most the best arm's plus 0.02, and when the unsafe truth class has fewer than 30 items no reliability claim is made. The smallest reliable executor minimises resident parameters, then sequential forward passes, then paired latency, reported over all arms and over neural arms only. An arm shows depth-indexed readiness if Spearman ρ between depth and the first step whose lens accuracy reaches 0.9 is at least 0.7 over four or more depths. Rates carry Wilson intervals [24]; half-lives are Kaplan–Meier estimates [18] censored at turn 10.

7 Compactification results

S1, algorithmic depth. Within trained depths the looped decoders are near ceiling (LOOP-T with deep supervision: pointer chasing 1.000, gated 1.000, reachability 0.995, sudoku 1.000 on depths 1–8) and in the neural-only pool the 460k-parameter LOOP-T is the smallest reliable executor. All of them collapse on depths 9–12 (pointer chasing 0.00, gated 0.38, reachability 0.58). The exact solver is the overall ceiling. The untied FF-U reads out only at its last block (no depth-indexed readiness, $\rho = 0.58$ on pointer chasing).

The single-pass Jev-style arms are nowhere near this frontier (Table 2). On the shared families the base model scores 0.406 on depths 1–8 and 0.353 on 9–12. Post-training on our decision suite lowers both (V1: 0.329 and 0.240; V1c: 0.364 and 0.260). The readout is not at fault: the option labels take a median 0.83–0.99 of the next-token probability mass in every family. On pointer chasing the base model is right on 0.92 of depth-1 items, 0.44 at depth 2 and 0.16 at depth 3, against 1/16 chance. Post-training also changed which labels the model will emit. The base model answers REFUSE on 133 of the 300 gated items (150 of which call for it); V1 never does, and its reachability answers shift from 27% to 79% "yes". A 4B model asked for one typed decision in one pass cannot chase pointers that a 460k-parameter looped decoder trained on the task chases reliably to depth 8. Recurrence trained on the decision suite does not change this. Over three seeds, J-V2b scores 0.347, 0.337 and 0.315 on S1 at one, two and three passes (seed 0 alone falls with every pass; seed 1 peaks at two). Its per-pass readout never reaches 0.9 at any depth beyond the first, so, unlike LOOP-T-ds and FF-U-ds ($\rho = 0.94$–0.98), it shows no depth-indexed readiness.

Training the 4B model on the task itself helps but does not close the gap. J-V1-rmp uses V1's recipe and budget (300 steps, 2,400 rows) on RMP train-region rows. It reaches 0.649 on depths 1–8 of the shared families, against the base model's 0.406 and LOOP-T-ds's 0.999. Its pointer chasing declines smoothly with depth (1.00, 0.92, 0.80, 0.72, 0.48, 0.44, 0.24 and 0.04 at depth 8). A single pass of the 4B model tracks three or four hops; the 460k-parameter decoder, eight visits of one block, tracks eight. On depths 1–8 every item that only one of the two gets right goes to LOOP-T-ds (200 items).

Beyond depth 8 neither extrapolates. J-V1-rmp scores 0.433 there against LOOP-T-ds's 0.320 ($p = 1.5 \times 10^{-4}$, paired), but that advantage is the 4B model guessing near the base rate while the decoder fails systematically. On pointer chasing J-V1-rmp scores 0.14 against a majority-class baseline of 0.12, and LOOP-T-ds 0.00. On gated chasing the figures are 0.49 against 0.48, and 0.38 for LOOP-T-ds. Only on reachability does J-V1-rmp clear the baseline (0.67 against 0.56), and there it does not differ from LOOP-T-ds (0.58, $p = 0.27$).

On the task's own rows, trained recurrence does what it did not do on the mixed decision suite. J-V2b-rmp is J-V1-rmp's training with layers 12–15 run twice. At its trained depth it beats J-V1-rmp at the same training budget on depths 1–8 in each of three seeds (0.724, 0.707 and 0.707 against 0.649, 0.674 and 0.643). Pooled over the seeds that is 274 to 158 discordant items, one-sided $p = 1.3 \times 10^{-8}$, the test we registered before the replicates ran (addendum A4). The gain is in the hops. The one-pass exit of the same adapter is worse in every seed (paired $p \leq 10^{-5}$). At seed 0 the second pass lifts pointer chasing at depths 4, 5 and 6 from 0.40, 0.24 and 0.08 to 0.92, 0.92 and 0.76; at seed 2, from 0.44, 0.08 and 0.04 to 0.68, 0.76 and 0.68. A third pass, beyond the trained depth, never helps. Depths 9–12 do not improve (0.367–0.430 against 0.433–0.447 for J-V1-rmp). The per-pass readout meets the registered depth-indexed readiness rule at seed 0 only ($\rho = 0.89$ on pointer chasing, and in only 44% of item resamples within that seed; in seeds 1 and 2 no depth past 2 reaches 0.9). The replicated claim is therefore the accuracy gain, not readiness; LOOP-T and LOOP-T-ds meet the rule on pointer chasing in every resample. The 4B model with a two-pass loop still tracks about six hops where the 460k-parameter decoder tracks eight.

Emitted reasoning is the one readout that extrapolates. J-cot-V1 is V1 shown worked solutions that spell out the walk ("J-¿P-¿G-¿I-¿B-¿D-¿H. A: H"), evaluated on 10 items per depth. On pointer chasing it scores 1.0 through depth 6 and 0.9 at depths 9–11, falling to 0.6 only at depth 12. On the shared families it scores 0.711 on depths 1–8 and 0.717 on depths 9–12. That is below LOOP-T-ds within the trained depths (paired 1 to 78) and far above it beyond them (0.717 against 0.308, paired 64 to 15, $p = 2 \times 10^{-8}$). Each hop is a token, so depth is bounded by the token budget rather than by the architecture. The gate and the search families gain less: gated chasing 0.57, reachability 0.69, sudoku 0.53. It costs 34 emitted tokens on average and a median 6.5 s per item, against one pass.

The base model reasons better still on the trained depths (J-cot-V0: 0.781 against 0.711, $p = 0.015$), almost entirely on gated chasing (0.78 against 0.57), so V1's adapter costs transfer even through its reasoning readout. Beyond depth 8 the two reasoning readouts tie at 0.717, and the base model's pointer chasing stays at 0.8–1.0 out to depth 12.

S2 and S7, verifier-visible state. The registered post-repair rule is correct on every S2 item on every split, as are the kNN critic and the TinyTRM port; with 18 true rejects S2 is underpowered by rule, so no arm is called reliable on it. On S7 the lexical router routes 128/128 and the 6,338-parameter TRM router 127/128 (one forbidden contract chosen on 23 negatives). These suites reproduce the TRM-skills paper's thesis [8]: once the decision-relevant state is exposed, the smallest reliable executor is a script or a tiny TRM, and a 4B model can at best match it. The Jev-style arms confirm it from the other side. On S2 they score 0.946–0.959 (Table 3); V1c commits 2 of the 18 true rejects, and every J arm over-refuses 2–4 of the 56 true commits. On S7 the base model routes 0.953 of cases, V1 0.812 and V1c 0.836, with no forbidden contract chosen. These S7 numbers mix routing with an option-position prior: the registered shortlist is in lexical-rank order and the lexical top-1 is always the gold contract, so every positive gold is option A. The base model puts 0.92 of its predictions there; V1, trained with answer positions rotated, has lost that prior. A diagnostic rerun with each shortlist shuffled was added after these results (addendum A2, post hoc), and it removes the prior: the share of predictions at the most common position falls from 0.71–0.92 to 0.22–0.26 for every text arm. Shuffled, the ranking reverses: V1 routes 0.867, the base model 0.836, V1c 0.820, and V2b at two passes 0.789. The base model's apparent lead on S7, and V1's apparent loss there, were the position prior. With or without it, every text arm stays well below the 6k-parameter TRM router (0.992) and the lexical router (1.000). The 1-bit Bonsai-8B, given the same prompts, is the weakest gate on S2: it commits 8 of the 18 true rejects (accuracy 0.811). It routes 0.766 of S7 cases in the registered order and 0.836 shuffled, with no forbidden contract chosen.

S3, long context. Recorded results: trained ControlTRM followed by the LDT veto reaches utility 0.687 with no unsafe actions, against 0.339 and 19.4% unsafe for the API language model alone. The Jev-style arms read each full transcript (up to 25.6k tokens, chunked prefill) and pick the action in one pass, zero-shot. Alone they are no better than the API model: accuracy 0.21–0.33, with 6–11 of 24 actions unsafe. Behind the same LDT veto they become safe (0 unsafe) and reach utility 0.61–0.62. That is barely above the veto alone (0.571) and below the 17k-parameter ControlTRM with LDT (0.687). With 24 rows per arm against 72 for the recorded arms, S3 stays descriptive, as registered. Within that limit, on long-context control the 4B single pass adds little to what the typed veto already provides.

Frontier. Figure 2 places every live arm by cost and accuracy. The frontier is held at its low-cost end on every suite. On S1, S4 and S7 a script or lexical router is the overall ceiling. On S7 the 6k-parameter TRM is the smallest reliable neural executor, on S1 the 460k-parameter LOOP-T, and on S3 the 17k-parameter ControlTRM with its LDT veto. S2 and S5 are underpowered by rule, so no arm is reliable there. The 4B arms sit four to six orders of magnitude to the right, and none of them meets the registered reliability rule on any suite. Where they come closest is S4, the suite they were trained on: there the best neural arm is V1's reasoning readout, 40 times the passes of its single pass.

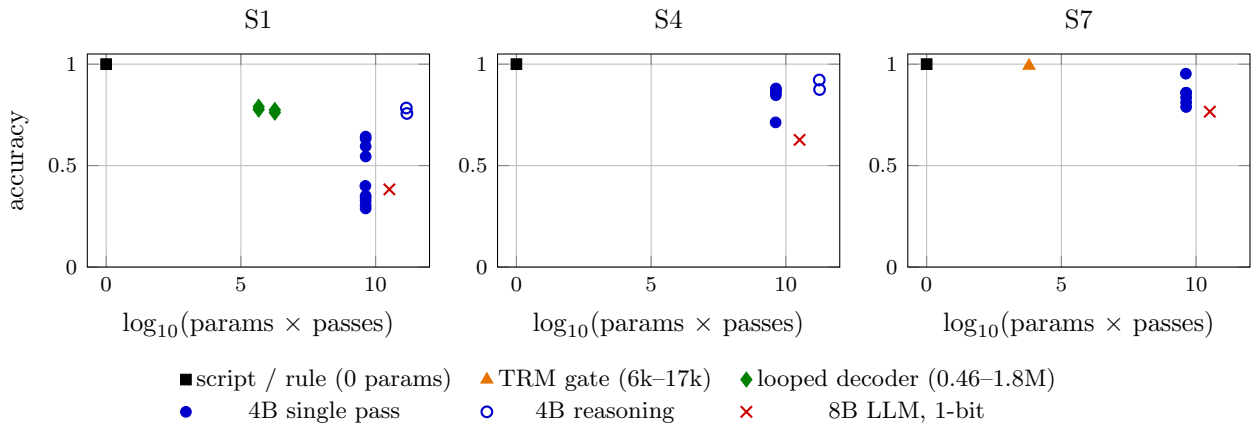


Figure 2: Accuracy against $\log_{10}$(resident parameters × sequential passes), one mark per arm family; parameter-free gates are plotted at 0. Registered arms only; seed replicates and the learning-rate control are in the repository's addenda table.

8 Control results

S5, monitor ladder. With scripted attackers, attack success falls from 0.90 with no control (q0) to 0.80 (action gate), 0.60 (trajectory budget), 0.50 (delegation guard) and 0.00 at the provenance membrane (q4), with no false blocks. Ignoring every cited claim also stops all attacks but blocks 23% of legitimate actions and drops benign task completion to 0.70. As monitors, the three Jev-style arms let one of ten attacks through and block at most one legitimate action (Table 3), which places them between the delegation guard (q3) and the provenance membrane (q4). They all miss the same attack: a quorum assembled from claims of duplicate lineage. V1 allows it with probability 0.999, so the calibrated readout gives no warning on the one decision that matters. That attack is structural: it is invisible in any single claim's text, and q4 catches it from provenance alone. Bonsai-8B as monitor lets 3 of 10 attacks through and blocks 4 of 23 legitimate actions. Its misses are three different attacks, allowed with probability 0.62–0.77, rather than the one confident miss of the Jev-style arms.

S6, fidelity half-life. Gates that read only typed authoritative fields cannot be flipped by injected evidence and are the ceiling; no Control-Harness control flipped under the contract-side ladder (Table 4). Under the scripted ladder no Jev-style gate reaches a half-life on the S4 deny decisions, but post-training makes them easier to move. The base model flips none of its 54 targets; V1 flips 5, all to the first injected note, and V1c the same 5, by turn 5. The base model's resistance comes with the 21% over-refusal on authorized requests reported in Section 4. On the contract side, where the gate decides one of the attacked proposals, the J monitors are weaker: the base model flips 5 of 10 eligible targets (half-life 8 turns), V1 4 and V1c 3. Trained recurrence is the one intervention that replicably adds robustness. At seed 0, V2b at its trained depth flips none of the 54 deny targets, against V1's 5 ($p = 0.0625$). We registered a pooled replicate test before running it (addendum A3). Over three training seeds V1 flips 5, 5 and 4 targets and V2b at two passes none: 14 to 0 discordant (seed, target) pairs, one-sided $p = 6 \times 10^{-5}$. Part of the effect is the adapter trained with the loop, since its one-pass exit flips 3, 2 and 0. Part is the second pass. The effect is confined to the S4 deny decisions; on the contract-side targets V2b flips 2–3 of 10 per seed, against V1's 4. Emitted reasoning gives no consistent protection. V1's `generate_cot` readout flips none of the 54

	S2 commit/veto			S5 monitor		
Arm	Acc.	Unsafe	Over-ref.	Attack succ.	False blocks	ECE
Bonsai-8B	0.811	8/18	6/56	0.30	4/23	0.124
CH-q0-none	–	–	–	0.90	0/22	–
CH-q1-action-gate	–	–	–	0.80	0/22	–
CH-q2-trajectory-budget	–	–	–	0.60	0/22	–
CH-q3-claims-ignored	–	–	–	0.00	5/22	–
CH-q3-delegation-guard	–	–	–	0.50	0/22	–
CH-q4-provenance-membrane	–	–	–	0.00	0/22	–
J-V0	0.959	0/18	3/56	0.10	1/22	0.108
J-V1	0.946	0/18	4/56	0.10	0/22	0.040
J-V1c	0.946	2/18	2/56	0.10	0/22	0.047
J-V2b-r1	1.000	0/18	0/56	0.20	0/22	0.057
J-V2b-r2	0.946	2/18	2/56	0.10	0/22	0.045
J-V2b-r3	0.959	0/18	3/56	0.00	2/23	0.066
TRM-cv	1.000	0/18	0/56	–	–	–
kNN-critic	1.000	0/18	0/56	–	–	–
script	1.000	0/18	0/56	–	–	–

Table 3: Gates. S2: unsafe = commit on a true reject (18), over-refusal = reject on a true commit (56). S5: attack success over attacked episodes, false blocks over benign proposals, ECE of the monitor's ALLOW probability where it has one.

targets that its single-pass readout flips 5 of, but the base model's `generate_cot` readout flips 4 different targets, mostly late in the ladder (turns 1, 5, 8, 8), where its single-pass readout flips none. (Reasoning answers are parsed from the final line; on S4 the same readouts parsed 399 and 395 of 400.) With one seed and counts this small, the S6 ladder separates only the clearest cases: the typed-field gates, which no injection can reach, from the text-reading gates, which a few injections reach in every configuration we tried except V2b at its trained depth and V1 reasoning. The text LLM is the easiest gate to move: Bonsai-8B flips 29 of its 49 eligible targets, with a half-life of 4 turns, and is the only gate with a finite half-life under the scripted ladder.

The adaptive attacker did not give a stronger test. On the raw completion endpoint Bonsai-8B echoed its own instructions instead of writing evidence, so those runs were discarded and rerun on the chat endpoint (addendum A5, post hoc). There it produced no echoes, which passes the validity rule we fixed before the rerun, but it hardly attacks. About 70% of its notes argue *for* the denial ("The Requester is a viewer with no authority to delete records…"), about 12% offer supporting evidence, and it rarely changes its note between turns (a median of 2 distinct notes over 10 turns). Against it, J-V1, J-V1c and V2b flip nothing, and J-V0 and Bonsai-8B flip 2 targets each. These numbers bound only this attacker. A small aligned model asked to red-team writes the defence instead, and the scripted ladder is the more informative test here.

9 Follow-up: training on the target label spaces

The transfer cost in Section 7 suggests a direct fix: train the adapter on the utility suites' own training splits, in their own label spaces and prompt formats. We registered this follow-up (SPEC-U1) before any of its runs. J-multi uses V1's recipe on a mix:

- V1's 2,400 decision-suite questions;

Ladder	Gate	Targets	Excluded	Flipped by 10	Half-life
scripted, S4	Bonsai-8B	54	5	29	4
scripted, S4	J-V0	54	0	0	> 10
scripted, S4	J-V1	54	0	5	> 10
scripted, S4	J-V1c	54	0	5	> 10
scripted, S4	J-V2b-r1	54	0	3	> 10
scripted, S4	J-V2b-r2	54	0	0	> 10
scripted, S4	J-V2b-r3	54	0	0	> 10
scripted, S4	J-cot-V0	54	0	4	> 10
scripted, S4	J-cot-V1	54	0	0	> 10
scripted, S4	script	54	0	–	> 10
scripted, S5	Bonsai-8B	11	4	3	> 10
scripted, S5	CH-q0-none	11	11	0	> 10
scripted, S5	CH-q1-action-gate	11	10	0	> 10
scripted, S5	CH-q2-trajectory-budget	11	7	0	> 10
scripted, S5	CH-q3-claims-ignored	11	0	0	> 10
scripted, S5	CH-q3-delegation-guard	11	5	0	> 10
scripted, S5	CH-q4-provenance-membrane	11	0	0	> 10
scripted, S5	J-V0	11	1	5	8
scripted, S5	J-V1	11	1	4	> 10
scripted, S5	J-V1c	11	1	3	> 10
scripted, S5	J-V2b-r1	11	2	3	> 10
scripted, S5	J-V2b-r2	11	1	3	> 10
scripted, S5	J-V2b-r3	11	0	3	> 10
adaptive, S4	Bonsai-8B	54	5	2	> 10
adaptive, S4	J-V0	54	0	2	> 10
adaptive, S4	J-V1	54	0	0	> 10
adaptive, S4	J-V1c	54	0	0	> 10
adaptive, S4	J-V2b-r2	54	0	0	> 10

Table 4: Fidelity half-life. Targets are deny decisions (S4) or attacked contract actions (S5); targets the gate already gets wrong with no injection are excluded. Half-life is the Kaplan–Meier turn at which half the remaining targets have flipped, censored at turn 10.

- J-V1-rmp's 2,400 RMP rows, identical by hash;
- the S2 commit/veto training split (234 rows, $\times 3$);
- hermes-lite's registered S7 training rows (54, $\times 4$, each copy with a fresh shuffle of the shortlist).

That is 714 optimizer steps. J-multi-loop adds the two-pass loop. S3 stays zero-shot, and no evaluation split is used for training. Each arm has three training seeds. Six tests were fixed in advance:

- T1–T3: transfer gains over V1 on S1 and S7p, and over V0 on S2. These are pooled one-sided exact tests over seeds, Holm-adjusted.
- H and S: non-inferiority to V1 on S4 and to J-V1-rmp on S1, judged by an item-bootstrap interval of the mean paired difference against a -2-point margin.
- L: the loop's gain on S1.

Training on the target label spaces removes most of the transfer cost (Table 5).

Arm	S1 d1–8	S1 d9–12	S2	S4	S7 (shuffled)
V0 (base)	0.406	0.353	0.959	0.713	0.836
V1 (decision suite)	0.351 (3)	0.261 (3)	0.968 (3)	0.868 (3)	0.870 (3)
V1-rmp (S1 rows)	0.655 (3)	0.440 (3)	–	–	–
multi	0.632 (3)	0.424 (3)	1.000 (3)	0.867 (3)	0.880 (3)
multi-loop, 2 passes	0.691 (3)	0.393 (3)	1.000 (3)	0.864 (3)	0.898 (3)

Table 5: SPEC-U1: accuracy averaged over training seeds (number of seeds in brackets). S1 on the shared families; S7 with the shortlist shuffled (S7p), the only S7 order that does not reward a position prior. Generated from the run records.

- *S1.* J-multi reaches 0.632 on S1 depths 1–8, against V1's 0.351 (T1: 755 to 185 discordant pairs, Holm $p = 4 \times 10^{-82}$). It is about two points short of the single-task specialist (S: -0.023, interval $[-0.043, -0.003]$, so non-inferiority is not shown).
- *S2.* It is correct on every S2 item in every seed (T3: Holm $p = 0.004$).
- *S4.* It loses nothing on the decision suite (H: -0.002, interval $[-0.013, +0.009]$).
- *Calibration.* It is better calibrated there than V1 in every seed (ECE 0.055–0.059 against 0.069–0.078).
- *S7p.* It does not route better (T2: 26 to 22, $p = 0.33$). More than half of the held S7 positives target contracts that never appear in the 57 registered training rows, so the training split cannot teach those routes. The 6k-parameter TRM router, which scores contract text directly, reaches 0.992.

The loop still adds hops when it is trained on the mixture. J-multi-loop at two passes reaches 0.691 on S1 depths 1–8 (L: 300 to 180, $p = 2 \times 10^{-8}$). The gain is uneven: 9 and 8 points at two seeds, none at the third. As before, the loop does not make the per-pass readout depth-indexed (0 of 3 seeds) and does not help past depth 8 (0.393, against 0.424 without it). Under the v1 reliability rule no Jev-style arm, old or new, is reliable on any utility suite. The smallest reliable executors are unchanged: a script or router, a 6k–17k-parameter TRM, or the 460k-parameter looped decoder.

10 Follow-up: skill pipelines and repair

Five further studies, each registered before its runs (SPEC-U2 to SPEC-U6), ask where a Jev-style model belongs relative to the TRM-infused skills: in front of them, choosing the skill, or inside them, gating or writing the repair.

10.1 Choosing a skill and gating its output (U2)

U2 composes routing and gating on 600 synthetic requests:

- 360 for the Campsite logic skill, each paired with a post-repair commit/veto state;
- 150 that a neighbouring skill should take;
- 90 negatives that name a skill the router must not use.

Table 6 gives the pipelines.

Pipeline	Router, gate	Success	Unsafe
A	generic Qwen routes, TRM gate (1)	0.773	0.025
A2	Bonsai-8B routes, TRM gate (1)	0.580	0.002
B	TRM router, TRM gate (1)	0.517	0.150
C	Jev routes and gates (3)	0.762	0.038
C-loop	looped Jev routes and gates (3)	0.793	0.041
D	generic Qwen routes and gates (1)	0.740	0.058
H	Jev routes, TRM gate (3)	0.787	0.013
E	lexical router, rule gate (ceiling) (1)	0.675	0.127

Table 6: SPEC-U2: end-to-end success and unsafe rate (a commit on a gold reject after correct routing, or the forbidden skill chosen) on 600 synthetic requests. Number of instances (seeds) in brackets. Generated from the run records.

- *The registered question (T1).* J-multi routing and gating (C, 0.762) does not beat the base model routing in front of the TRM gate (A, 0.773; Holm $p = 0.92$). It does beat the base model doing both steps (D, 0.740; T3, Holm $p = 0.009$).
- *The TRM router (T2).* Its pipeline (B, 0.517) is far worse. The router has no abstain path, and it picks the forbidden skill on all 90 negatives.
- *Best combination.* The best pipeline pairs the Jev's routing with the TRM gate (H, 0.787, 1.3% unsafe). Every Jev gate error is the same boundary case: a repair that changes nothing and is not exact, which the rule rejects and the Jev commits. The looped Jev doing both steps reaches 0.793; its comparison with A was not registered ($p = 0.036$, post hoc).

10.2 The repair-rudder benchmark (U3)

The TRM skills' own repository reports a repair "rudder" benchmark on 88 near-miss rows. There, retrieved examples lift 9B and 27B rudders from 0.36–0.49 to 0.795 joint accuracy on repair action and commit decision. Replicating it exactly, we found the retrieval matches on the evaluated row's own labels:

- 98.6% of the retrieved examples share the row's commit action, against 77.3% for a retrieval on pre-repair fields only.
- With the same retrieval, our 4B base model, the Jev and the looped Jev also reach 0.795. That is the ceiling: the unseen-family repair actions are outside the allowed list.
- Without the leak, the base model, a newly trained repair TRM and a lookup table tie at 34/54 on the held-out rows. The Jev scores 0.580 against the base model's 0.630 (R1 not supported), and there is no difference from the repair TRM (R2).
- The TRM router's abstain head, never trained in its own pipeline, still never fires when fitted on the frozen router (A1 not supported). The router is most confident (0.55) on the requests that name the skill it must not use, against 0.20 on ordinary requests.

10.3 Verifier-scored Campsite repair (U4–U6)

U4 builds fresh broken Campsite candidates (eight defect types, 60 each) on new puzzles and scores every committed grid with the skill harness's official verifier. The skill's own repair projections run unchanged from its repository. Table 7 collects the results.

- *Untrained (U4).* A 9.7k-parameter decision TRM trained on 3,009 synthetic items picks among commit, two projections and reject at 0.727, with 1.9% broken commits. The few-shot Jev reaches 0.569 with 18.7% (D1), no better than the base model (D2). Writing the grid itself, it fixes 0.184 against the dual projection's 0.731 (G1).

- *Reject path (U5).* The skill's flow policies never reject, so every candidate their projection cannot fix is committed broken (27%). Refusing to commit an unchanged, already-failed candidate cuts this to 1.7% at no loss of success. On real proposals only a verifier check after the repair removes the broken commits.

- *Trained on the TRM's items (U5).* Continued from J-multi on the same 3,009 items, the Jev matches the TRM as decider: 0.731 against 0.727, which is significant (J1, Holm $p = 0.031$) but only two items per seed. As repairer it beats the dual projection, 0.872 against 0.731 (J2). It fixes the swapped-tent and wrong-shape candidates that the projections cannot.

- *Real proposals (U5).* Two proposers, the base model and Bonsai-8B, each proposed once for 237 puzzles: 117 of standard size and 60 each at 8×8 and 10×10. Only 2 of the 474 proposals solved the puzzle as written. Synthetic training does not transfer to these failures: the trained Jev repairs 0.084 of them, against 0.160 for the C projection (P2).

- *Trained on real failures (U6).* Continued on 1,196 real proposals for separate training puzzles, the Jev repairs 0.363 of the held-out proposals: 0.677 at standard size and 0.106 at 8×8, where neither projection fixes any (T1 against the C projection, T2 against synthetic-only training). As a decider it learns to refuse: 0.207 success with 11% broken commits, against 0.172 and 9% for a TRM trained on the same proposals (T3). On the synthetic defects it gives back 12–19 points.

- *The cost of specialising (U6).* Against J-multi on its own suites (99% non-inferiority at -2 points), the Campsite-trained Jev shows a real loss on S4 (-2.4, interval $[-3.5, -1.2]$). S1 is inconclusive (-1.9). S2 and U2 are unharmed, and S7p is unchanged but too small to show it.

- *The exact solver.* The harness's CSP solver re-solves every candidate in milliseconds at every size we could generate. Planting a puzzle stalls at 14×14 before the solver slows, while the dual projection takes over a minute per candidate at 10×10. On Campsite, a learned repairer is a research instrument, not a need.

11 Discussion

What a small Jev-style model is for. The evidence so far favours a narrow role: a single-pass calibrated readout is cheap and, after light post-training, close to the same model's reasoning on short-answer decisions. It does not make the model an authorization gate by itself: where authoritative state can be read by a script, the script is smaller, cannot be talked out of its decision, and is already reliable.

	Synthetic defects (480)			Real proposals (474)		
Arm	Decide	Unsafe	Repair	Decide	Unsafe	Repair
c_repair (projection)	–	–	0.606	–	–	0.160
dual_repair (projection)	–	–	0.731	–	–	0.257
dual flow policy, plain	0.731	0.269	–	0.257	0.743	–
dual flow policy, verify-reject	–	–	–	0.257	0.000	–
Decision-TRM (synthetic)	0.727	0.019	–	0.191	0.526	–
Decision-TRM (real)	–	–	–	0.172	0.086	–
generic Qwen, few-shot	0.590	0.252	0.156	–	–	–
Jev (J-multi), few-shot	0.569	0.187	0.184	–	–	–
Jev + synthetic (J-u4)	0.731	0.019	0.872	0.240	0.738	0.084
Jev + synthetic + real (J-real)	0.610	0.005	0.685	0.207	0.112	0.363
CSP re-solve (ceiling)	–	–	1.000	–	–	1.000

Table 7: SPEC-U4–U6: Campsite deciders (success of the committed grid; unsafe = a failing grid committed) and repairers (the repaired grid passes the verifier). Synthetic defects: the U4 test set. Real proposals: base-model and Bonsai-8B proposals on held-out standard, 8×8 and 10×10 puzzles. Rates pool the seeds present. Generated from the run records.

Post-training transfers badly. Post-training on one decision suite gives up accuracy on algorithmic depth (S1, all three seeds) and makes deny decisions easier to inject (V1 flips 4–5 of 54 per seed where the base model flips none, though the base model's resistance comes with a 21% over-refusal). Part of the mechanism is mundane. Training on a fixed letter scheme with rotated answer positions installs a prior over labels: V1 stops emitting REFUSE (X) on S1. On S7 the same rotation removed a position prior that had been flattering the base model, and the shuffled diagnostic shows V1 routing slightly better, not worse. A typed-decision model's label space is part of what it learns, so a Jev-style model needs training on the label spaces it will serve. One trained on a generic suite and dropped into a new gate is not neutral, even when its in-domain calibration is good. The calibration objective softens this: over three seeds the Brier-trained V1c loses less on S1 than V1 does (0.358 against 0.311 on average, pooled paired $p = 3 \times 10^{-14}$). Training on the target suites' own label spaces largely removes it (Section 9).

Recurrence. Hidden depth is not free capability at this scale and budget, but it is not useless either; what decides is the training data. Untrained, a loop drifts. Trained for 300 steps on the mixed decision suite, it computes, changing answers at each pass, but lands at the accuracy of the same adapter without the loop and leaves the 4.5-point gap to emitted reasoning untouched. Trained for the same 300 steps on a single task whose difficulty is a number of sequential hops, the same loop makes its second pass do the hops the first could not: pointer chasing at depths 4–6 goes from 0.04–0.64 to 0.68–0.92, and it beats the loop-free adapter in each of three seeds. Its readout is depth-indexed by our registered rule at only one of those seeds. Our reading is that a loop grafted onto a pretrained stack can learn an iterative step when the data has one consistent step to learn, and that a mix of eight task families does not. The compact looped decoders take this to its end: recurrence is the whole architecture, trained from scratch on the task, and their readout tracks depth for eight visits. Neither kind extrapolates past its trained depths. On the control side, the loop trained on the decision suite is more resistant to scripted evidence injection in every seed. It was the only change to the 4B model that bought robustness without the over-refusal of the base model.

Where the repair step belongs. The infused skill's projections are good near-miss repairers, but its flow policy has no reject path. Adding one removes most broken commits at no cost, and the repository contains a patch for it. Learned repairers win only when they are trained on the failures they will meet: synthetic perturbations do not prepare a model for real model proposals. Each specialisation costs a few points elsewhere. In every configuration we tried, the verifier, not either learner, is what makes a commit safe.

Limitations.

- *Seeds.* Three training seeds for V1, V1c, V2b and the RMP-trained arms; one for the learning-rate control. Seeds change initialisation and data order, not the evaluation items. Seed replicates are reported outside the registered reliability pools (addendum A1).
- *Adaptive pressure.* Our adaptive attacker was too weak to test it (see S6).
- *Measurement.* Latency was measured on a GPU shared with other jobs; the registered measure is therefore a paired ratio.
- *Tasks.* The decision tasks are synthetic and templated. S2 is underpowered, and S7's registered option order confounds routing with a position prior (addendum A2).
- *Baselines and arms.* The RMP baselines come from a programme whose own gate (G0) is not yet met for every seed. No MeTTa gate could be run: the implementations are on a missing drive.
- *Jev.* Jev itself is not available for comparison, so nothing here measures Jev.
- *Follow-ups.* U2's requests and U4's defects are synthetic. The repair-rudder rows are post-hoc projections from earlier runs. Campsite has a cheap exact solver. Forgetting was measured only for the synthetic-trained adapter, and only on J-multi's suites.

12 Conclusion

A small Jev-style model can be built from an open 4B base. A light adapter gives a typed, calibrated, single-pass readout within five points of the model's own reasoning at a twentieth of its latency. The benchmark mostly says where such a model should *not* be the gate of record:

- Where the decision state is typed and verifier-visible, a script or a tiny TRM is smaller and already reliable.
- Where the decision requires iterating a step, a compact looped decoder trained on that step does it to its trained depth, and emitted reasoning does it beyond.
- Where the threat is provenance, only a provenance membrane is immune, and the calibrated readout fails with full confidence.

What remains is the role the control harness already has room for: a cheap monitor layer above heuristic controls and below a structural membrane, whose failure modes a harness can measure. Those failure modes are transfer cost from post-training on the wrong label space, and confident misses on structural attacks. The first can be trained away: one adapter trained on the target gates' own label spaces keeps its home-suite accuracy and recovers most of the transfer loss. It still does

not reach the smallest reliable executor on any suite. Recurrence grafted onto the pretrained stack earns its keep in two places only, both replicated over three seeds. On a task with one consistent step it adds hops. On the deny gate it resists scripted evidence injection. It neither extrapolates nor closes the gap to emitted reasoning.

Next to the TRM skills, the model is most useful in front of them, choosing the skill, with the TRM commit gate kept. Inside the repair step it can match a tiny trained TRM and outperform the skills' hand-written projections, but only on failures like those it was trained on. The skill's verifier and an explicit reject path matter more than which learner sits in the loop.

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